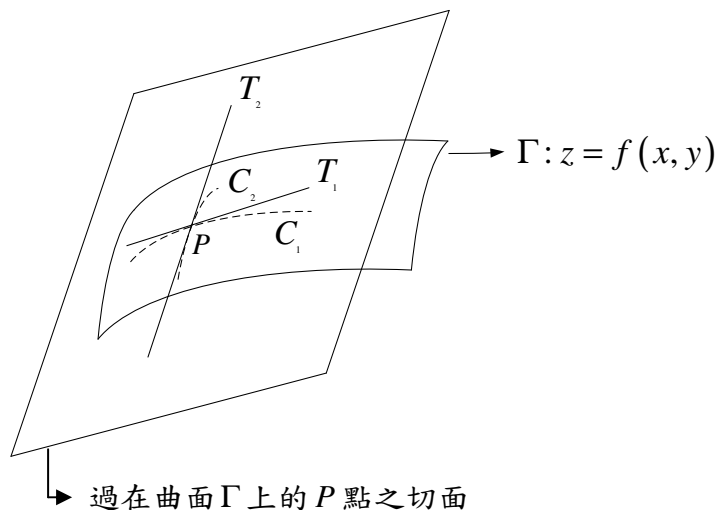


§14.4 Tangent Planes and Linear Approximation

Homework : 3,5,11,15,19,23,27,31,33,39,42

(1)切面方程式



設此切面方程式為

$$A(x-x_0)+B(y-y_0)+(z-z_0)=0 \quad (1)$$

令 $x=x_0$, 則

$B(y-y_0)+(z-z_0)=0$ 為過 P 點和曲線 $C_1: z=f(x_0, y)$ 相切的直線方程式 T_1 .

$$\Rightarrow f_y(x_0, y_0) = T_1 \text{ 的斜率} = -B.$$

$$\text{同理 } f_x(x_0, y_0) = T_2 \text{ 的斜率} = -A$$

公式 : (i) 切面的法向量為

$$\langle A, B, -1 \rangle = \langle f_x(x_0, y_0), f_y(x_0, y_0), -1 \rangle$$

(ii) 切面方程式

$$z-z_0 = f_x(x_0, y_0)(x-x_0) + f_y(x_0, y_0)(y-y_0)$$

Example 1 : $z = 2x^2 + y^2$. 過 $P(1,1,3)$, 找過 P 點的切面方程式 .

Solution :

$$f_x(x, y) = 4x \Rightarrow f_x(1, 1) = 4 .$$

$$f_y(x, y) = 2y \Rightarrow f_y(1, 1) = 2 .$$

$$\text{Tangent plane : } 4(x-1) + 2(y-1) = z-3 .$$

(2) Linear approximation of $z = f(x, y)$ at (a, b) .

(以切平面來逼近曲面) .

$$z = f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b) .$$

Example 2 : $z = \sqrt{x^2 + y^2}$ at $(3, 4)$. Find $z(3.01, 3.99)$.

Solution :

$$f_x(x, y) = \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}}(2x) \Rightarrow f_x(3, 4) = \frac{3}{5}$$

$$f_y(x, y) = \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}}(2y) \Rightarrow f_y(3, 4) = \frac{4}{5}$$

$$\Rightarrow z = 5 + \frac{3}{5}(0.01) + \frac{4}{5}(-0.01) = 4.998 \approx \sqrt{(3.01)^2 + (3.99)^2} .$$

(3) Differentiable of f at (a, b) :

Definition : If $\Delta z = f(a + \Delta x, b + \Delta y) - f(a, b)$

$$= f_x(a, b)\Delta x + f_y(a, b)\Delta y + \underbrace{\varepsilon_1\Delta x + \varepsilon_2\Delta y}_{\text{誤差}} ,$$

then $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$ (切面 $\approx$ 曲面)

Example 3 : $f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & , (x, y) \neq (0, 0). \\ 0 & , (x, y) = (0, 0). \end{cases}$

• $f_x(0, 0) = f_y(0, 0) = 0$.

• Tangent plane through $(0, 0) : z = 0$.

• But $f(x, y) = \frac{1}{2}$ whenever $x = y \neq 0$.

$\Rightarrow f$ is not differentiable at (a, b) .

Theorem : f is differentiable at (a, b) provided that

(i) f_x, f_y exist near (a, b) .

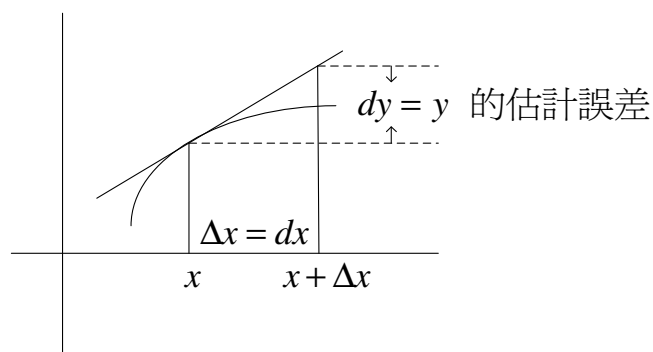
(ii) f_x, f_y are continuous at (a, b) .

Remark : Differentiable $\Rightarrow$ Continuous.

(4) Differentiables

Recall :

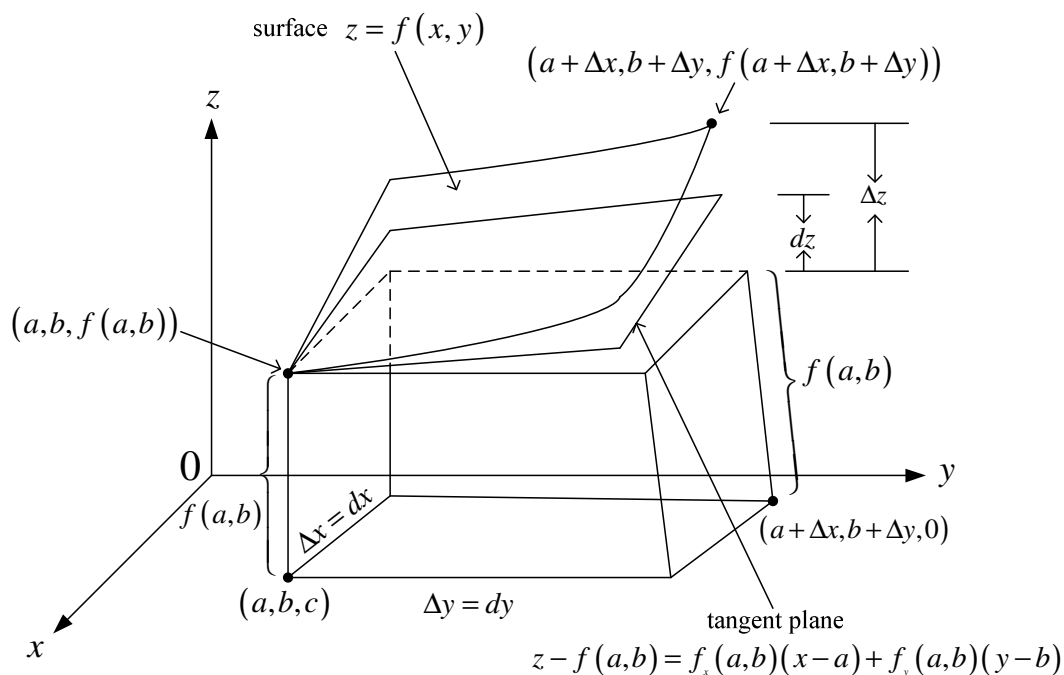
• 1-D :



$\Delta y = f(x + \Delta x) - f(x)$ = 實際誤差.

$\Delta y \approx dy = f'(x)dx$ (切線代曲線得到的誤差).

• 2-D :



$$\Delta z = f(a + \Delta x, b + \Delta y) - f(a, b) = \text{實際誤差}$$

dz = 估計誤差(利用切平面代曲面得到的誤差)

$$= f_x(a, b)dx + f_y(a, b)dy.$$

$$\text{公式 } \boxed{\Delta z \approx dz = f_x(a, b)dx + f_y(a, b)dy}$$

Example 4 : $z = f(x, y) = x^2 + 3xy - y^2$

(i) Find dz (ii) Use differentials to estimate $f(2.05, 2.96)$.

Solution :

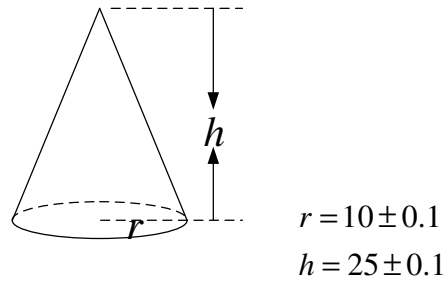
$$(i) \quad dz = (2x + 3y)dx + (3x - 2y)dy$$

$$(ii) \quad dz = (2.2 + 3.3)(0.05) + (3.2 - 2.3)(-0.04) = 0.65$$

$$(\text{令 } a = 2, b = 3 \Rightarrow \Delta x = 0.05, \Delta y = -0.04).$$

$$\text{註 } \Delta z = f(2.05, 2.96) - f(2, 3) = 0.6449.$$

Example 5 :



Use differentials to estimate the maximum error in the calculated volume of the cone.

Solution :

$$V = \frac{1}{3}\pi r^2 h$$
$$dV = \frac{2}{3}\pi h r dr + \frac{1}{3}\pi r^2 dh$$
$$= 20\pi$$